

Subject Description Form

Subject Code	EIE3109
Subject Title	Mobile Systems and Application Development
Credit Value	3
Level	3
Pre-requisite / Co-requisite/ Exclusion	EIE3320 Object-Oriented Design and Programming
Objectives	This course aims at providing students with an understanding of the real-time embedded and mobile systems, and the techniques essential to the design and implementation of mobile applications.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the structure of real-time operating systems for modern mobile computer systems. 2. Understand the programming techniques and tools for developing software that is run in modern mobile computer systems 3. Apply the knowledge to develop practical applications for modern real-time mobile computer systems. <u>Category B: Attributes for all-roundedness</u> 4. understand the creative process when designing solutions to a problem
Subject Synopsis/ Indicative Syllabus	1. <u>Introduction</u> Introduction to Embedded Systems – embedded real-time systems, embedded programming and program models, real-time operating system (RTOS). Introduction to Mobile Systems and Mobile Application Development 2. <u>iOS Application Development</u> Introduction to iOS – system architecture, development environment (Xcode), MVC and MVVM architecture. Application development with Xcode. 3. <u>Android Application Development</u> Introduction to Android OS – development environment (Android Studio), Android application basic (activity, service, content provider, broadcast receiver, intent resolution). Application development with Android Studio 4. <u>Introduction to Cross-platform Development</u> Comparison between native and cross-platform development. Application development with cross-platform development environment.
Teaching/Learning Methodology	Lectures: The subject matters will be delivered through lectures. Students will be engaged in the lectures through Q&A, discussions and specially designed classroom activities.

	Tutorials: During tutorials, students will work on/discuss some chosen topics in small group. This will help strengthen the knowledge taught in lectures. Laboratory and assignments: During laboratory exercises, students will perform hands-on tasks to practice what they have learned. They will evaluate performance of systems and design solutions to problems. The assignments will help students to review the knowledge taught in class. While lectures and tutorials will help to achieve the professional outcomes, the open-ended questions in laboratory exercises and assignments will provide the chance to students to exercise their creativity in problem solving.					
Assessment Methods in Alignment with Intended Subject Learning Outcomes	Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)			
			1	2	3	4
	1. Continuous Assessment (total: 100%)					
	• Homework and assignments	30%	✓	✓	✓	✓
	• Tests/Quizzes	50%	✓	✓	✓	✓
	• Laboratory exercises	20%			✓	✓
	Total	100%				
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assignment, homework and laboratory exercises will require students to apply what they have learnt to solve problems. There will be open-ended questions that allow students to exercise their creativity in making design. Tests: They assess students’ achievement of the learning outcomes more rigorously.						
Student Study Effort Expected	Class contact (time-tabled):					
	• Lecture					24 Hours
	• Tutorial/Laboratory/Practice Classes					15 hours
	Other student study effort:					
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes					36 Hours
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing					30 Hours
Total student study effort:					105 Hours	
Reading List and References	Reference Books: 1. Raj Kamai, <i>Embedded Systems: Architecture, Programming and Design</i>, 3rd ed., McGraw-Hill, 2015. 2. Sahar, Ahmad ; Clayton, Craig, <i>IOS 16 Programming for Beginners : Kickstart Your IOS App Development Journey with a Hands-On Guide to Swift 5. 7 and Xcode 14</i>, 7th Edition. , Birmingham: Packt Publishing, Limited 2022.					

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| | <ol style="list-style-type: none">3. Wei-Meng Lee, <i>Beginning Swift programming</i>, John Wiley & Sons 2015.4. Clayton, Craig, <i>SwiftUI Projects</i>, Packt Publishing 2020.5. J. F. DiMarzio, <i>Beginning Android programming with Android studio</i>, Fourth edition, Wrox, a Wiley brand 2017. |
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